PUSL3122 HCI, Computer graphics and visualisation

Topics covered last in lecture 2

- **What data** needs to be gathered
  - How to plan and run a successful data gathering program.
  - How to plan and run an interview, design a simple questionnaire and plan and carry out an observation.
- What do I **do with my data**?
  - Difference between qualitative and quantitative data and analysis; how to present findings
  - Software packages available to help with analysis
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapters 8 and 9.

Chapter 8

DATA GATHERING

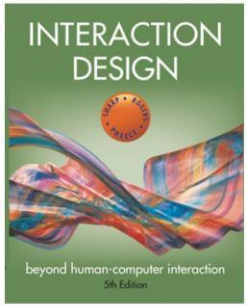
Chapter 9
Data Analysis, Interpretation, and Presentation

Topics covered this lecture (lecture 3) …

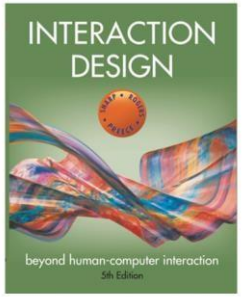
- **Evaluation**: key concepts, terms, different types of evaluation methods. Different evaluation methods, Hybrid approach is best.

- **Practice**: how to do usability testing, outline the basics of experimental design, how to do field studies

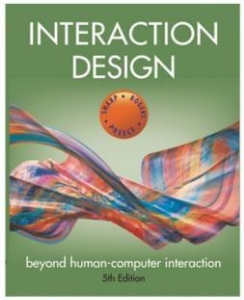
- **Inspection**: evaluation without users



Chapter 14
INTRODUCING
EVALUATION



Chapter 15
Evaluation Studies



Chapter 16
Evaluation

Evaluation: Why, what, where, and when

- Iterative design and evaluation is a **continuous process** that examines:
  - **Why**: To check users' requirements and confirm that users can use the product and that they like it
  - **What**: A conceptual model, early and subsequent prototypes of a new system, more complete prototypes, and a prototype to compare with competitors' products
  - **Where**: In natural, in-the-wild, and laboratory settings
  - **When**: Throughout design; finished products can be evaluated to collect information to inform new products

Bruce Tognazzini - why evaluation needed

- "*Iterative design, with its repeating cycle of design and testing, is the only validated methodology in existence that will consistently produce successful results. If you don't have user-testing as an integral part of your design process you are going to throw buckets of money down the drain*."
- See AskTog.com for topical discussions about design and evaluation – it makes for great reading

Types of evaluation

- **Controlled** settings that directly involve users
  - Most popular
  - for example, usability and research labs
- **Natural** settings involving users
  - Gaining in popularity
  - for instance, online communities and products that are used in public places
  - Often there is little or no control over what users do, especially in in-the-wild settings
- **Any setting** that doesn't directly involve users
  - Not used as much
  - for example, consultants and researchers critique the prototypes, and may predict and model how successful they will be when used by users

Living labs

- People's use of technology in their everyday lives can be evaluated in living labs

https://www.youtube.com/watch?v=LxJ9iuB4mTE&ab_channel=TWISTProject Accessed December 2020

- Such evaluations are too difficult to do in a usability lab
- An early example was the Aware Home that was embedded with a complex network of sensors and audio/video recording devices (Abowd et al., 2000)

https://www.youtube.com/watch?v=dmDoSOv85o0&ab_channel=CambridgeUniversity

Abowd, G., and Mynatt, E. (2000) Charting Past, Present, and Future Research in Ubiquitous Computing. ACM Transactions on Computer–Human Interaction, 7(1), 29–58.

What is a Living Lab?

Inclusion Living Lab
00:00 – 04:20 (ALL)

https://www.youtube.com/watch?v=Y7K3-kEKmQE&ab_channel=LuxMagazine

Living labs

- More recent examples include **EU funded research** project where whole **blocks and cities** that house hundreds of people
- Example, Verma et al., research in Switzerland (2017)

Verma, H., Alavi, H. S., and Lalanne, D. (2017) Studying Space Use: Bringing HCI Tools to Architectural Projects. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2017). ACM, Press, New York, pp. 3856–3866.

https://ec.europa.eu/research/openvision/pdf/rise/cities_as_living_labs.pdf
Accessed December 2020

Evaluation case studies

- **Highly** recommend you look at these
  - A **classic experimental investigation** into the physiological responses of players of a computer game
  - An ethnographic study of visitors at the **Royal Highland show** in which participants are directed and tracked using a mobile phone app https://petras-iot.org/update/chatbot-ethnographer-gathers-data-at-the-royal-highland-show/
  - Crowdsourcing in which the **opinions and reactions of volunteers** (for example, from the crowd) inform technology evaluation

The language of evaluation

- Analytics
- Analytical evaluation
- Biases
- Controlled experiment
- Crowdsourcing
- Ecological validity
- Expert review or criticism
- Field study
- Formative evaluation
- Heuristic evaluation
- Informed consent form
- In the wild evaluation
- Living laboratory
- Predictive evaluation
- Reliability
- Scope
- Summative evaluation
- Usability laboratory
- User studies
- Usability testing
- Users or participants
- Validity

Participants' rights and getting their consent

- Participants need to be told **why** the evaluation is being done, **what** they will be asked to do and **informed about their rights**
- Informed consent forms **provide** this information and act as a **contract** between participants and researchers
- The design of the informed consent form, the evaluation process, data analysis, and data storage methods are typically **approved by a high authority**, such as the Institutional Review Board
  - In our case this is carried out by members of the University Ethics committee
  - See here for full details https://www.plymouth.ac.uk/research/governance/research-ethics-policy (accessed December 2020)

HRA Ethical Approval Animation

00:00 – 02:14 ALL

https://www.youtube.com/watch?v=Wo2oeS6buTs&ab_channel=KeepingYouHealthy

Things to consider when interpreting data

- **Reliability**:
  - Does the method produce the **same** results on **separate** occasions?
- **Validity**
  - Does the method **measure** what it is **intended** to measure?
- **Ecological validity**
  - Does the **environment** of the evaluation **distort** the results?
- **Biases**
  - Are there **biases** that distort the results?
- **Scope**
  - How **generalizable** are the results?

Usability testing

- Involves recording **performance of typical users** doing typical tasks
- **Controlled** settings
- **Users** (or participants) are observed and timed
- **Data is recorded** on video, and key presses are logged
- The data is used to **calculate performance times** and to identify and explain errors
- **User satisfaction is evaluated** using questionnaires and interviews
- Field observations **may** be used to provide contextual understanding

Usability lab with observers watching a user and assistant

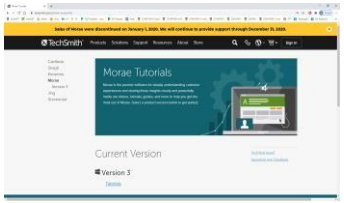
- 2 staff
- 2 users
- High cost?

Tobii Glasses Mobile Eye-Tracking System

Dalton et al., 2015, p.3891. Reproduced with permission of ACM Publications.

Wixon, D., and Wilson, C. (1997) The Usability Engineering Framework for Product Design and Evaluation. In M. G. Helander, T. K. Landauer and P. V. Prabju (eds) Handbook of Human–Computer Interaction. Elsevier, Amsterdam, pp. 653–688.

Smartlook software



- Morae was an awesome product (discontinued 2020)
- Smartlook is free to use
  https://www.youtube.com/watch?v=bXPMewY8AOI

Testing the iPad usability

- First study was conducted **quickly** in two cities: Fremont, CA and Chicago, IL
- Tests had to be done quickly, as **information was needed by third-party app developers**
- Also needed to be done **secretly** so that the competition was not aware of the study before the iPad was launched
- Seven participants with over three months experience with iPhones

Testing the iPad usability (continued)

- Signed an **informed consent** form explaining:
  - **What** the participant would be asked to do
  - The **length of time** needed for the study
  - The **compensation** that would be offered for participating
  - Participants' **right to withdraw** from the study at any time
  - A promise that the person's **identity** would not be disclosed
  - An agreement that the data collected would be **confidential** and available to only the evaluators
- Participants were asked to **explore** the iPad
- Next, they were asked to **perform randomly-assigned specified tasks**

Tasks used in the iPad evaluation

App or website	Task
iBook	Download a free copy of *Alice's Adventures in Wonderland* and read through the first few pages.
Craigslist	Find some free mulch for your garden.
Time Magazine	Browse through the magazine and find the best pictures of the week.
Epicurious	You want to make an apple pie tonight. Find a recipe and see what you need to buy in order to prepare it.
Kayak	You are planning a trip to Death Valley in May this year. Find a hotel located in the park or close to the park.

Source: iPad App and Website Usability Study. Used courtesy of the Neilsen Norman Group.

Adapted from Budiu, R., and Nielsen, J. (2010) *Usability of iPad Apps and Websites. First Research Findings*. Nielsen Norman Group. Downloaded from www.nngroup.com/reports/mobile/ipad/ (retrieved August 2010). https://www.nngroup.com/articles/ipad-usability-first-findings/ (accessed December 2020).

Problems and actions

NOTE: Another study was done a year later to examine this and other issues that there was insufficient time to address in the first study

- Examples of problems detected:
  - **Access**ing the Web was difficult
  - Lack of **affordance** and **feedback**
  - Getting **lost** in an application
  - Knowing where to tap
- Actions by evaluators:
  - Reported to developers
  - Made available to public on Neilsen Norman Group.
- Accessibility for **all** users is important (but not easy to achieve)
- Study did not address how iPad would be used in people's everyday lives

Usability testing conditions

- Usability **lab** or **other controlled** space
- Emphasis on:
  - Selecting **representative users**
  - Developing **representative tasks**
- **5-10 users** typically selected
- Tasks usually around **30 minutes**
  - It's important that tasks are **sufficiently complex and thorough !!!!!**
- Test conditions are the **same for every participant**
- Informed consent form explains procedures and deals with ethical issues

How many participants is enough?*

- The number is a **practical** issue but like most things it comes down to **time available** and **resources available.** It will depend on:
    - Schedule for testing
    - Availability of participants
    - Cost of running tests
- Typically, 5-10 participants
    - It's a bit controversial these days to **state this number** but I believe it's a sensible estimate for usability in practice
- Some experts argue that testing should continue until no new insights are gained
    - I disagree – even big players like Apple don't have that **endless time and money**

Usability testing and Experiments

- Usability testing is **applied experimentation**
- Developers check that the system is usable by the intended user population by **collecting data** about participants' **performance** on prescribed **tasks**
- **Experiments test hypotheses** to discover new knowledge by investigating the relationship between two or more variables

Usability testing Vs Research

Usability Testing

- Improve **products**
- **Few** participants
- Results inform design
- Usually not completely replicable
- Conditions controlled as much as possible
- Procedure planned
- Results reported **to developers**

Experiments for Research

- Discover **knowledge**
- **Many** participants
- Results validated statistically
- Must be **replicable**
- Strongly controlled conditions
- Experimental design
- Scientific report to **scientific community**

Different, same, matched participant design

Design	Advantages	Disadvantages
Different	No order effects	Many subjects and individual differences a problem
Same	Few individuals, no individual differences	Counter-balancing needed because of ordering effects
Matched	Same as different participants, but individual differences reduced	Cannot be sure of perfect matching on all differences

Field studies

- Field studies are done in **natural settings**
- "**In the wild**" is a term for **prototypes** being used freely in natural settings
- Seek to understand what users do **naturally** and how technology impacts them
- Field studies are used in product design to:
  - Identify opportunities for **new** technology
  - Determine **design** requirements
  - Decide how best to **introduce new technology**
  - **Evaluate** technology in use

EXAMPLE:
A field study of a pain-monitoring device

- Monitoring patients' pain is a **known challenge** for physicians
- Goal of the study was to **evaluate the use of a pain-monitoring device** for use after surgery
- **Painpad** is a keypad device
  - It was usability tested **extensively in the lab before** being brought into two hospitals
  - Goal was to understand how Painpad was used in the natural environment and as part of routines in two UK hospitals.
  - How pain-monitoring **differed** with Painpad (current system vs new innovation)

Painpad

A tangible device for inpatient self-logging of pain

Source: Price, B., Kelly, R., Mehta, V., McCormick, C., Ahmed, H., and Pearce, O. (2018) Feel My Pain; Design and Evaluation of Painpad, a Tangible Device for Supporting Inpatient Self-Logging of Pain.
In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI'18).* ACM, New York, NY, Paper 169, 13 pages.

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Data collection and participants

- Two studies in two hospitals involving 54 people
- 13 males, 41 females
- **Privacy** was an important concern
- Hospital **stay** ranged from **1-7 days**, mean and median **age 64.6, 64.5**
- Patients **given Painpad after surgery** and **prompted** to report pain levels every two hours
- Nurses also collected scores
- All data was entered into charts
- Patients in one hospital were given a user-satisfaction survey when they left
- Also rated Painpad on a 1-5 Likert scale

Data analysis and presentation

- **Three types of data** were collected:
  - **Satisfaction** with Painpad was based on questionnaire responses
  - Patients' **compliance** with the two-hour routine
  - How data collected from **Painpad compared with data collected by nurses**
- Data showed:
  - Satisfaction with Painpad **4.63 on Likert** scale
  - Patience compliance was mixed: some liked it while others disliked or **didn't notice** the prompts
  - Patients **recorded more** scores with Painpad than through the nurses

Does all **evaluation** have to involve **users**?

Inspections

- Several kinds
- **Experts** use their knowledge of users and technology to **review software usability**
- **Expert critiques** can be formal or informal
- **Heuristic evaluation** is a review guided by a set of heuristics
- **Walkthroughs** involve stepping through a pre-planned scenario noting potential problems

Heuristic evaluation

- Developed by **Jacob Nielsen** in the early 1990s
- Based on heuristics distilled from **an empirical analysis of 249 usability problems**
- These heuristics have been **revised** for current technology by Nielsen and others for Mobile devices, Wearables, Virtual worlds & Social media
- Design guidelines form a basis for developing heuristics

Revised version (2014) of heuristics

- **Recognition** rather than recall
- **Flexibility** and **efficiency** of use
- **Aesthetic** and **minimalist** design
- Help users **recognize, diagnose, recover** from errors
- Help and documentation
- Visibility of **system status**
- Match between system and real world
- User control and freedom
- Consistency and standards
- Error prevention

Examples

- **10 Usability Heuristics Applied to Complex Applications** — Examples of the heuristics applied to complex and domain-specific software applications.

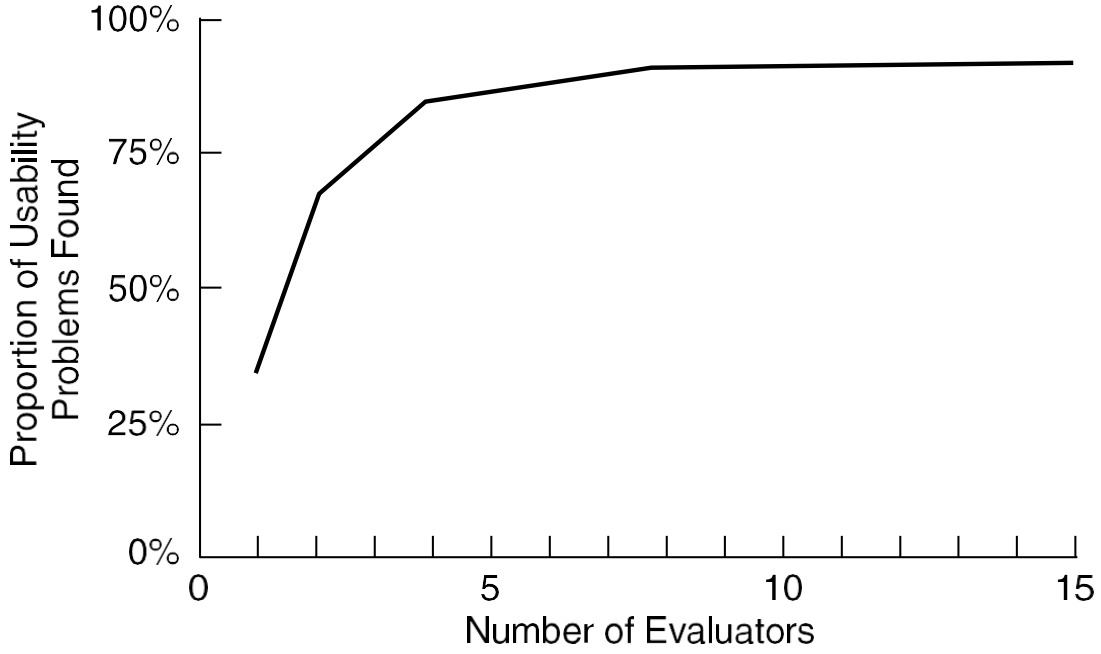
- **10 Usability Heuristics Applied to Virtual Reality** — See the heuristics applied to 3D virtual environments.

- **10 Usability Heuristics Applied to Video Games** — Great examples of the 10 heuristics in highly interactive and highly visual user interfaces that have an entertainment purpose.

- **10 Usability Heuristics Applied to Everyday Life** (Just for fun)

Number of evaluators and problems

Curve showing the proportion of usability problems in an interface found by heuristic evaluation using **different numbers of evaluators**



Source: Nielsen, J., and Mack, R.L. (eds) (1994) *Usability Inspection Methods*. John Wiley & Sons Inc., New York.

Use courtesy of Wiley.

Number of evaluators

- Nielsen suggests that on average **five** evaluators identify **75-80 percent of usability problems**

- Cockton and Woolrych point out that the number of users needed to find 75-80 percent of usability problems depends on the **context** and **nature** of the task problems.

Cockton, G., and Woolrych, A. (2001) Understanding Inspection Methods: Lessons from an Assessment of Heuristic Evaluation. In A.Blandford and J.Vanderdonckt (eds), *People and Computers XV*. Springer-Verlag, Berlin, pp. 171–191.

Heuristics for websites focus on key criteria

- **Clarity**
- Minimize unnecessary **complexity** and **cognitive load**
- Provide users with **context**
- Promote positive and **pleasurable** user experience

Budd, A. (2007) *Web Heuristics*. Available at: www.andybudd.com/archives/2007/01/heuristics_for_modern_web_application_development/ (accessed December 2020).

Source: Budd, 2007

Turning guidelines and rules into heuristics

- Ask questions like the following:
  - “Does the application include a visible title page, section or site?
  - Does the user always know where they are located?
  - Does the user always know what the system or application is doing?
  - Are the links clearly defined?
  - Can all actions be visualized directly (i.e., no other actions are required)?”

Granollers p. 62

Granollers, T. (2018) Usability Evaluation with Heuristics, Beyond Nielsen's list. In *Proceedings of the 8th International Conference on Advances in Human Computer Interaction*. pp. 60–65.

Evaluating for accessibility using Guidelines

- Web Content Accessibility Guidelines (WCAG)

  Lane, N. D., and Georgiev, P. (2015). Can Deep Learning Revolutionize Mobile Sensing? In *Proceedings of the 16th International Workshop on Mobile Computing Systems and Application*. ACM, New York, NY, pp. 117–122.

- **Guidelines** can be used as **heuristics** for evaluating websites
- Governments and large corporations have to make their websites accessible by law (Universities too – have you seen the icons on the DLE?)
- Four key concepts:
  - **Perceivable**
  - **Operable**
  - **Understandable**
  - **Robust**

Source: WCAG 2.1 at a Glance.accessed December 2020

The three questions

- Will the **correct action** be sufficiently **evident** to the user?
- Will the **user notice** that the **correct action** is available?
- Will the user **associate and interpret** the response from the **action** correctly?

As the experts work through the scenario, they note problems

Web Analytics

- A form of **interaction logging** that **analyzes users' activities** on website
- Designers **use the analysis to improve** their designs

- When designs don't meet users' needs, they will not return to the site
  - They become one-time users

A/B Testing

- A l**arge-scale** experiment
- Offers **another way** to evaluate a website, application of app running on a mobile device
- Often used for **evaluating changes in design** on social media applications
- **Compares how two groups of users perform** on two versions of a design
- Can involve **thousands of users**
- May **create ethical dilemmas** if users don't know they are part of the test
- **Care** is needed to ensure that other issues are not affecting users' behaviour

Predictive models

- Provide a way of **evaluating** products or designs **without** directly involving **users**
- **Less expensive** than user testing
- Usefulness **limited** to systems with predictable tasks, for example, voicemail systems, smartphones, and dedicated mobile devices
- Based on expert error-free behavior

Fitts' Law (1954)

- **Fitts' Law** predicts that the **time to point at an object** using a device is a **function** of the **distance** from the target object and the **object's size**
- The **further away** and the **smaller the object**, the longer the time to locate it and point to it
- It is particularly useful for determining **where** on a screen to position an object
- Fitts' Law is useful for evaluating systems for which the **time** to locate an object **is important**, for example, smartphones, handhelds, and mobile devices

Thank you